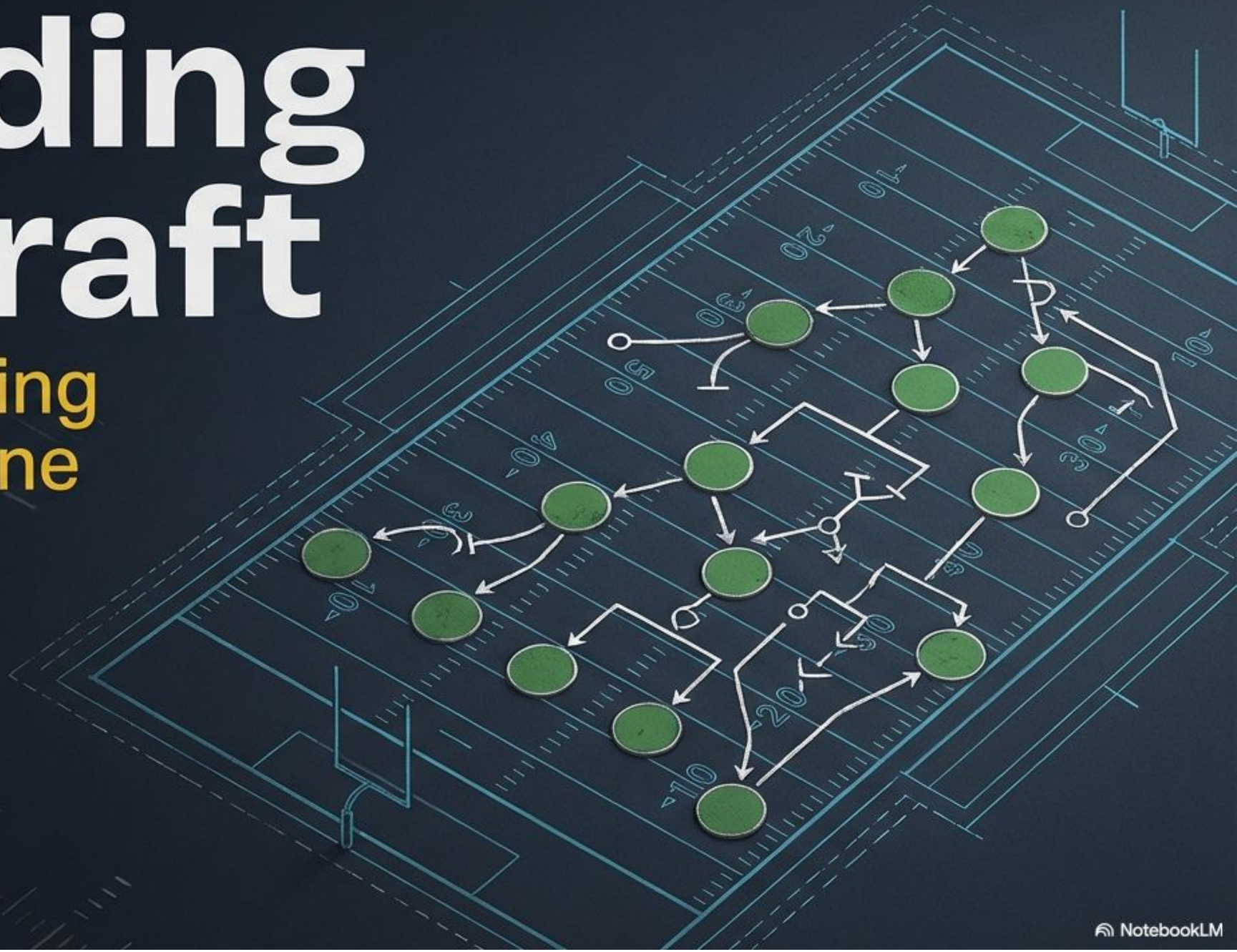
Decoding the Draft

Machine Learning the NFL Combine



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Quantifying Athletic Potential

The Old Paradigm

The Data Reality

Player ID	Position	Height	Weight	Speed	Bench	Target Variable
4258	LB	74.5	225	4.45	28	
3912	WR	72.0	190	4.32	15	
5021	OL	77.0	310	5.20	32	
...	...	...	...	...	...	
...	...	...	...	...	...	

1 (Drafted)

0 (Not Drafted)

The Goal: Determine if machine learning algorithms can predict college player draft success via binary classification of objective physical measurements.

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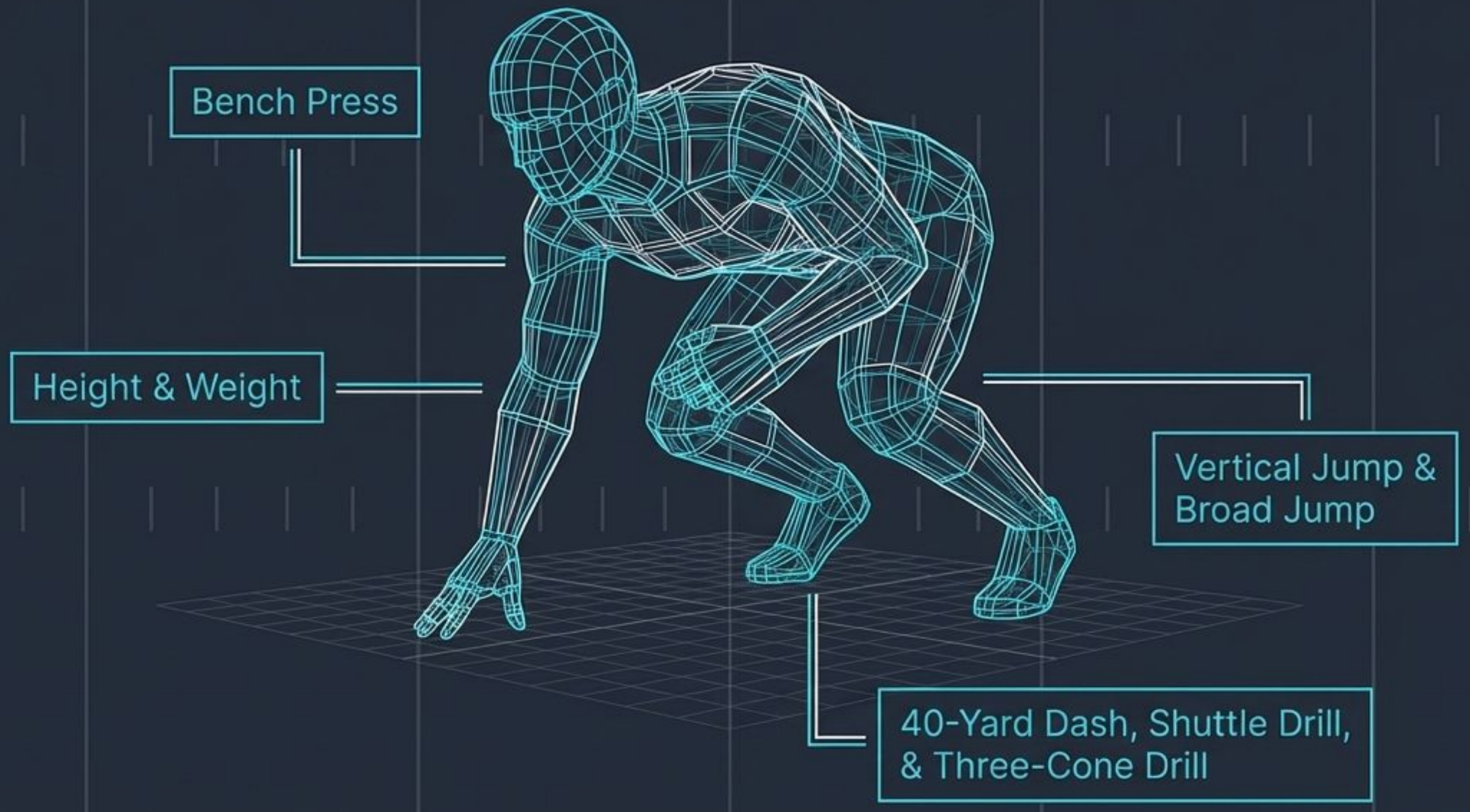
The Anatomy of a Combine Data Point

Dataset Scope:	2009–2019
Sample Size:	3,477 distinct players



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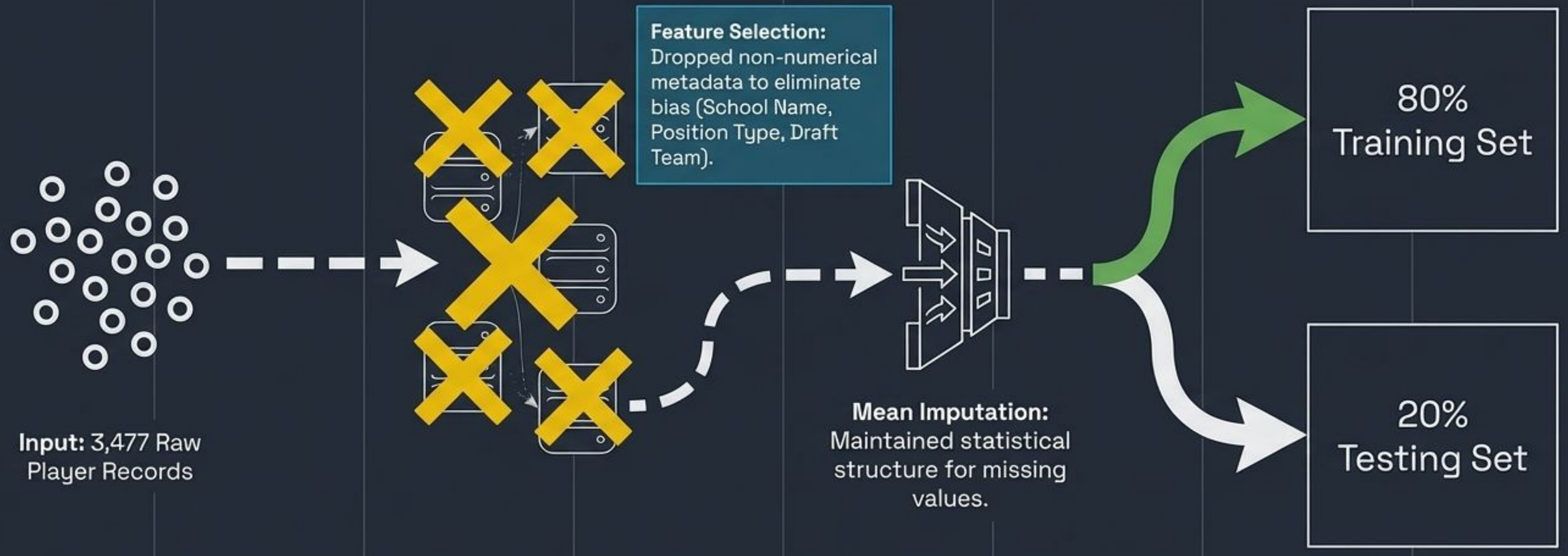
Executing the Preprocessing Playbook



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Evaluating the Algorithm Arena

Four models were trained on the identical 80/20 split to establish a fair performance baseline and evaluate the impact of algorithmic complexity.

Algorithm	Methodology Type	Accuracy Score	Performance
Logistic Regression	Linear Baseline	72.98%	72.98%
Support Vector Machine (SVM)	Vector Space	67.30%	67.30%
HistGradientBoosting	Ensemble Tree	80.40%	80.40%
XGBoost	Ensemble Tree	81.00%	81.00%

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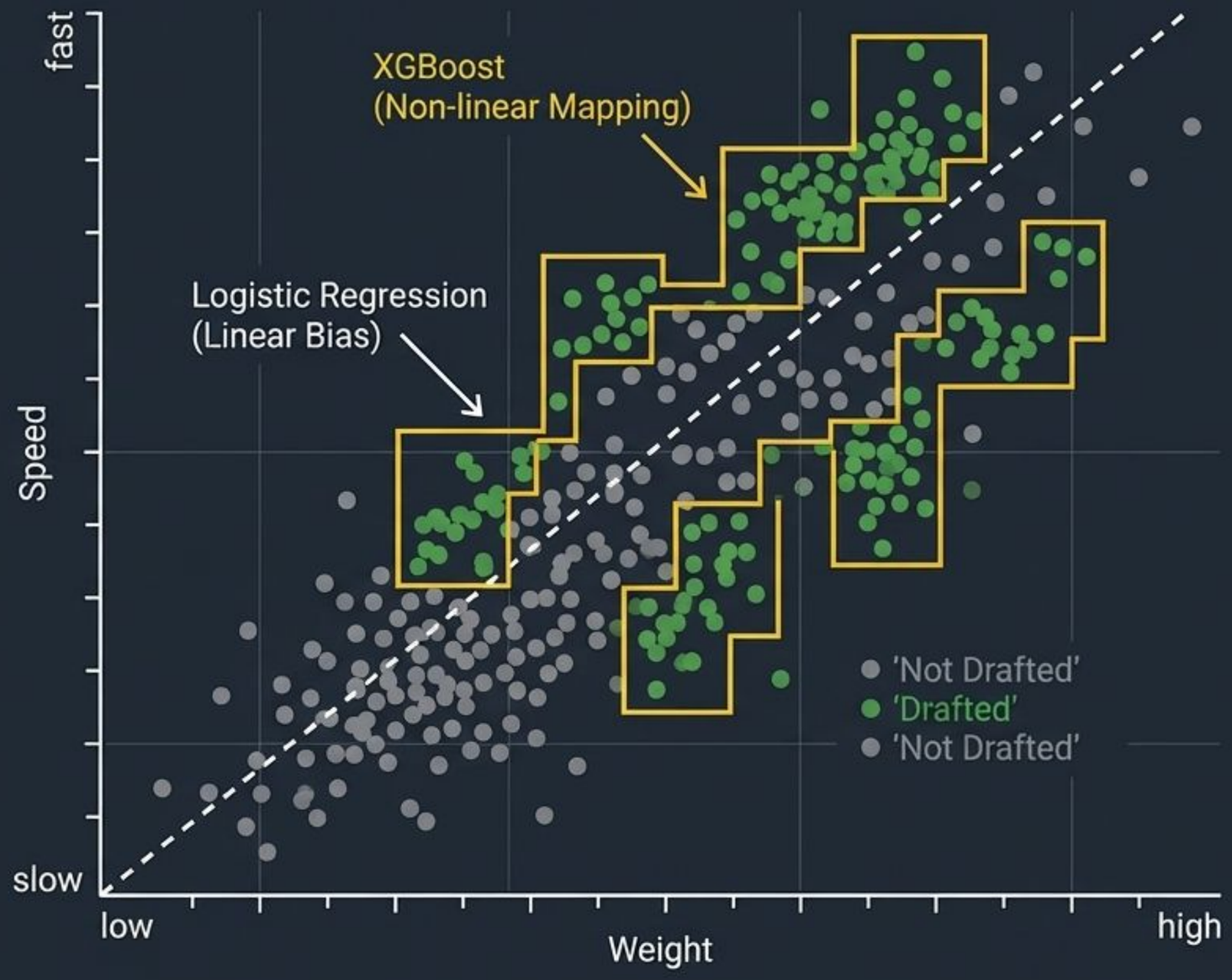
“Capturing Latent Physical Profiles” in Space Grotesk

By removing ‘position type’ during preprocessing, the models were forced to evaluate raw physical ratios.

Linear models struggle because physical value is relative: a heavy player with a fast 40-yard dash represents a completely different athletic profile than a light player with a fast 40.

XGBoost achieved peak accuracy because ensemble trees inherently excel at mapping complex, non-linear relationships.



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The Objective Value of the Combine in Space Grotesk

The Mathematical Proof

NFL Combine performance metrics contain significant, measurable information that directly correlates to draft probability.

The Algorithmic Edge

Relying on baseline linear evaluations leaves actionable data on the table. Advanced machine learning models can successfully identify the non-linear athletic thresholds that separate a drafted player from the rest of the pack.

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